MCO-1D Build Measurement Sheet

As-built dimensions needed to synchronize the wave library software with the constructed hardware. Fill in each blank after the corresponding mechanism is built and measured. Units as noted.

1. Sector Wheel / Flap Drive (highest priority — gates everything else)

Measurement	Why it's needed	Current placeholder	Your value
Actual total sector wheel arc as cut/printed (deg)	Sets the real mechanical ceiling for flap amplitude	200° (target)	
Sector wheel pivot-to-rim radius (mm)	Used in all clearance and wingtip-position calculations	45 mm (estimated)	
Capstan / small drum radius (mm)	Sets rope travel and reduction ratio	15 mm (estimated)	
Actual reduction ratio achieved (large wheel : small drum)	Torque, speed, and frequency calculations all depend on this	7:1 (estimated)	
Root mechanism housing — width / depth / height (mm)	Replaces the conservative 30 mm spherical assumption used in collision checks	30 mm radius sphere (placeholder)	
ODrive vel_limit value actually saved on the unit	Confirms the frequency-ceiling fix was applied	should be 150	
Measured vel_limit ceiling frequency achieved on the bench (Hz)	Calibrates FREQ_LIMIT_MAX_HZ to reality	3.0 Hz (target, unconfirmed)	

2. Wing Root / Staircase Geometry

Measurement	Why it's needed	Current placeholder	Your value
Actual Z-step built between adjacent wing pairs (mm)	Replaces estimated STAIRCASE_Z_STEP_MM	100 mm	
Actual root chord at the pivot, as tapered (mm)	Replaces ROOT_CHORD_AT_PIVOT_MM	40 mm	
Span fraction where full chord is reached (%)	Replaces ROOT_TAPER_FRACTION	15%	
Measured clearance between adjacent wings at full deflection (mm)	Bench measurement — validates or corrects the entire collision model	calculated only, never measured	
Actual X-stagger, root-to-root fore-aft (mm)	Confirms STAGGER_MM matches the design	138 mm	

3. Pitch Servo Installation (per wing — at minimum one representative wing)

Measurement	Why it's needed	Current placeholder	Your value
Measured PWM pulse width at true mechanical neutral, 0° AoA (microseconds)	Sets PITCH_SERVO_NEUTRAL_OFFSET_US per wing	0 (untested)	
Measured gain correction — commanded angle vs. actual angle via digital protractor	Sets PITCH_SERVO_GAIN per wing	1.0 (untested)	

Measurement	Why it's needed	Current placeholder	Your value
Actual supply voltage delivered to pitch servos under load (V)	Confirms PITCH_SERVO_VOLTAGE_V and all speed/torque figures derived from it	7.4 V (assumed)	
Confirmed servo model actually installed	Confirms whether TCM-1001 or A11CLS V3 spec values apply	TCM-1001 assumed	

4. Sweep Servo / Stow Mechanism

Measurement	Why it's needed	Current placeholder	Your value
Confirmed sweep servo model and actual travel range (deg)	Confirms SWEEP_LIMIT_MIN_DEG / SWEEP_LIMIT_MAX_DEG	0°-210° (DS3225 assumed)	
Measured torque required to hold sweep against airflow, at actual fuselage geometry (kg·cm)	Validates whether the installed servo is adequate	not yet measured	

5. Wing Physical Properties (per wing, or per pair if uniform)

Measurement	Why it's needed	Current placeholder	Your value
Actual built mass per wing (g)	Feeds inertia and torque calculations throughout the library	estimated, 52 g used in some calcs	
Actual chord as built (mm)	Cross-check against CHORD_MM	165 mm	
Actual semi-spans as built — W1, W2, W3, W4 (mm)	Cross-check against SEMI_SPANS_MM	772 / 749 / 735 / 712 mm	
Measured wing thickness at thickest point (mm)	Cross-check against the 12%c S1223 assumption	19.8 mm (calculated)	
Total aircraft mass at this build stage (kg)	Cross-check against MTOW_KG	3.448 kg	

6. Capstan Rope / Drive Cord

Measurement	Why it's needed	Current placeholder	Your value
Confirmed rope diameter actually installed (mm)	Confirms whether the 0.75 mm switch was made	1.8 mm, unless changed	
Measured peak motor current at 3 Hz on the bench (A)	Validates the torque/current model	calculated only	

Once complete, return this sheet (or the values from it) and every constant above will be updated in the wave library together, with the full collision-safety report and servo self-test re-run against the real numbers.